

Q 1. Consider a grammar $G = (V, T, P, S)$ where-

- $V = \{ S \}$
 - $T = \{ a, b \}$
 - $P = \{ S \rightarrow aSbS, S \rightarrow bSaS, S \rightarrow \epsilon \}$
 - $S = \{ S \}$
-
- This grammar is an example of a context free grammar.
 - It generates the strings having equal number of a's and b's.

Q2. Language: of all strings having many 0's, defined over $\{0\}$

Regular Expression: a^+

Context Free Grammar: $S \Rightarrow 0S \mid 0$

Q.3

Language: of all strings having many 0's or no zero, defined over $\{0\}$

Regular Expression: a^*

Context Free Grammar: $S \Rightarrow 0S \mid \epsilon$

Q4.

Language: of all strings having exactly one 0's or exactly one 1, defined over $\{0,1\}$

Regular Expression: $0+1$

Context Free Grammar: $S \Rightarrow 0 \mid 1$

Q 5. Language: of all strings having many 0's or many 1's or no zero or no one, defined over $\{0,1\}$

Regular Expression: $(0+1)^*$

Context Free Grammar: $S \Rightarrow 0S \mid 1S \mid \epsilon$

5. Build the CFG for the language of all those strings having 0 a's or many a's?

$S \Rightarrow aS \mid \epsilon$

Now we can read multiple a's from this CFG.

For example;

To read 0 a's from $S \Rightarrow aS \mid \epsilon$

$S \Rightarrow aS \mid \epsilon$

$S \Rightarrow \epsilon$

To read 1 a from $S \Rightarrow aS \mid \epsilon$

$S \Rightarrow aS$

$S \Rightarrow a \epsilon$

$S \Rightarrow a$

To read 2 a from $S \Rightarrow aS \mid \epsilon$

$S \Rightarrow aS$

$S \Rightarrow aaS$

$S \Rightarrow a \epsilon$

$S \Rightarrow a$

and similarly, we can read many a's from this CFG.

Q.7 Examples of CFG of odd Length strings

To read a;

CFG:

$S \rightarrow aX \mid bX$

$X \rightarrow aS \mid bS \mid \epsilon$

Path to Read the String

$S \Rightarrow aX$ //Read first a.

$X \Rightarrow \epsilon$ //Read first a ϵ .

finally, we can read it as a.

Congratulation, the string can be read and the machine can be stoped after successfully completing its task.

To read b;

CFG:

$S \rightarrow aX \mid bX$

$X \rightarrow aS \mid bS \mid \epsilon$

$S \Rightarrow bX$ //Read first b.

$X \Rightarrow \epsilon$ //Read first b ϵ .

finally, we can read it as b.

Q. 6

CFG $\{N,T,P,S\}$ be

$N = \{S\}$, $T = \{a, b\}$, Starting symbol = S , $P = S \rightarrow SS \mid aSb \mid \epsilon$

One derivation from the above CFG is "abaabb"

$S \rightarrow SS \rightarrow aSbS \rightarrow abS \rightarrow abaSb \rightarrow abaaSbb \rightarrow abaabb$

Q.7. Verify the the string aabbabba for the CFG given by is valid,

1. $S \rightarrow aB \mid bA$
2. $A \rightarrow a \mid aS \mid bAA$
3. $B \rightarrow b \mid bS \mid aBB$

Q.8 Show the derivation tree for string "aabbbb" with the following grammar.

1. $S \rightarrow AB \mid \epsilon$
2. $A \rightarrow aB$
3. $B \rightarrow Sb$

Q. 9 Grammar: $S \rightarrow ASB \mid cA \rightarrow \epsilon \mid aAB \rightarrow \epsilon \mid bB$

Derivation Sequences for acb

$S \Rightarrow ASB \Rightarrow aASB \Rightarrow aSB \Rightarrow acB \Rightarrow acbB \Rightarrow acb$

$S \Rightarrow ASB \Rightarrow ASbB \Rightarrow ASb \Rightarrow aAcB \Rightarrow aAcB \Rightarrow acb$

$S \Rightarrow ASB \Rightarrow AcB \Rightarrow aAcB \Rightarrow aAcBb \Rightarrow acbB \Rightarrow acb$

Problem-10:

Consider the grammar-

$S \rightarrow bB \mid aA$

$A \rightarrow b \mid bS \mid aAA$

$B \rightarrow a \mid aS \mid bBB$

For the string $w = bbaababa$, find-

1. Leftmost derivation
2. Rightmost derivation

3. Parse Tree

Solution-

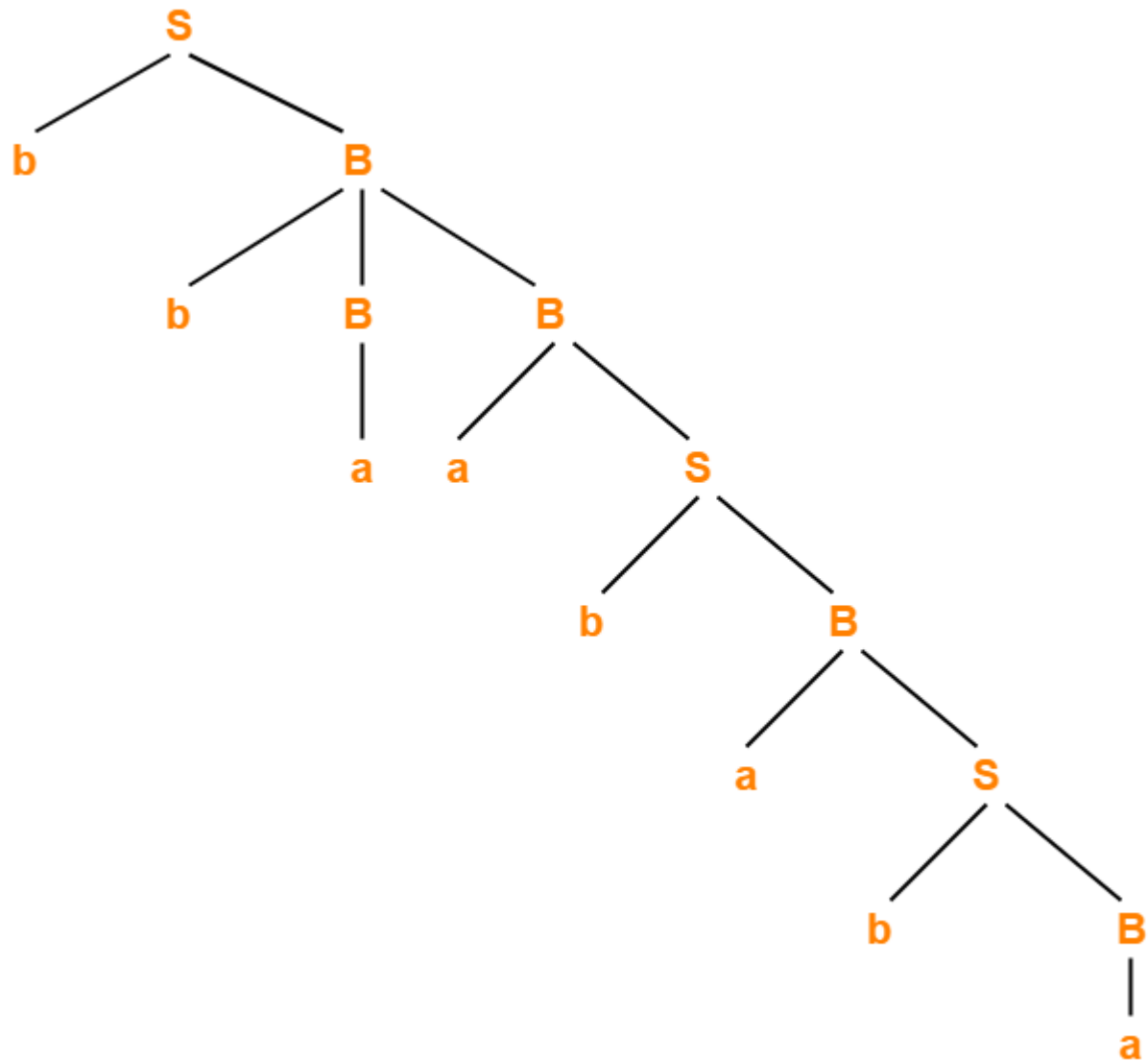
1. Leftmost Derivation-

$S \rightarrow bB$
 $\rightarrow bbBB$ (Using $B \rightarrow bBB$)
 $\rightarrow bbaB$ (Using $B \rightarrow a$)
 $\rightarrow bbaaS$ (Using $B \rightarrow aS$)
 $\rightarrow bbaabB$ (Using $S \rightarrow bB$)
 $\rightarrow bbaabaS$ (Using $B \rightarrow aS$)
 $\rightarrow bbaababB$ (Using $S \rightarrow bB$)
 $\rightarrow bbaababa$ (Using $B \rightarrow a$)

2. Rightmost Derivation-

$S \rightarrow bB$
 $\rightarrow bbBB$ (Using $B \rightarrow bBB$)
 $\rightarrow bbBaS$ (Using $B \rightarrow aS$)
 $\rightarrow bbBabB$ (Using $S \rightarrow bB$)
 $\rightarrow bbBabaS$ (Using $B \rightarrow aS$)
 $\rightarrow bbBababB$ (Using $S \rightarrow bB$)
 $\rightarrow bbBababa$ (Using $B \rightarrow a$)
 $\rightarrow bbaababa$ (Using $B \rightarrow a$)

3. Parse Tree-



Parse Tree

- Whether we consider the leftmost derivation or rightmost derivation, we get the above parse tree.
- The reason is given grammar is unambiguous.

Problem-11:

Consider the grammar-

$$S \rightarrow A1B$$

$$A \rightarrow 0A / \epsilon$$

$$B \rightarrow 0B / 1B / \epsilon$$

For the string $w = 00101$, find-

1. Leftmost derivation
2. Rightmost derivation
3. Parse Tree

Solution-

1. Leftmost Derivation-

$$S \rightarrow A1B$$

$$\rightarrow 0A1B \text{ (Using } A \rightarrow 0A \text{)}$$

$$\rightarrow 00A1B \text{ (Using } A \rightarrow 0A \text{)}$$

$$\rightarrow 001B \text{ (Using } A \rightarrow \epsilon \text{)}$$

$$\rightarrow 0010B \text{ (Using } B \rightarrow 0B \text{)}$$

$$\rightarrow 00101B \text{ (Using } B \rightarrow 1B \text{)}$$

$$\rightarrow 00101 \text{ (Using } B \rightarrow \epsilon \text{)}$$

2. Rightmost Derivation-

$$S \rightarrow A1B$$

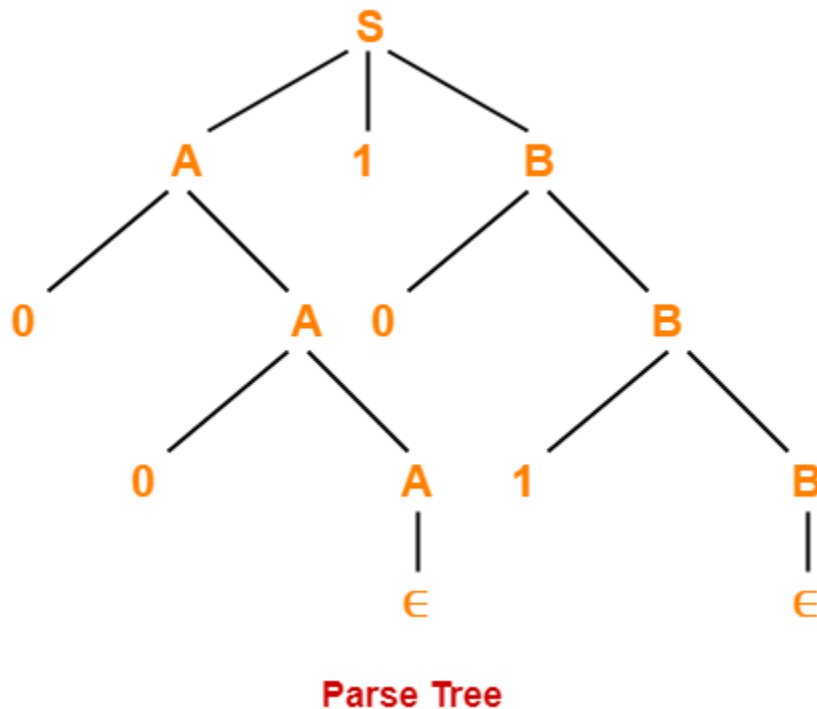
$$\rightarrow A10B \text{ (Using } B \rightarrow 0B \text{)}$$

$$\rightarrow A101B \text{ (Using } B \rightarrow 1B \text{)}$$

$$\rightarrow A101 \text{ (Using } B \rightarrow \epsilon \text{)}$$

$\rightarrow 0A101$ (Using $A \rightarrow 0A$)
 $\rightarrow 00A101$ (Using $A \rightarrow 0A$)
 $\rightarrow 00101$ (Using $A \rightarrow \epsilon$)

3. Parse Tree-



- Whether we consider the leftmost derivation or rightmost derivation, we get the above parse tree.
- The reason is given grammar is unambiguous.

Q. 12

Problem

Convert the following CFG into CNF

$S \rightarrow ASA \mid aB$, $A \rightarrow B \mid S$, $B \rightarrow b \mid \epsilon$

Solution

(1) Since **S** appears in R.H.S, we add a new state **S₀** and **S₀ → S** is added to the production set and it becomes –

$S_0 \rightarrow S, S \rightarrow ASA \mid aB, A \rightarrow B \mid S, B \rightarrow b \mid \epsilon$

(2) Now we will remove the null productions –

$B \rightarrow \epsilon$ and $A \rightarrow \epsilon$

After removing $B \rightarrow \epsilon$, the production set becomes –

$S_0 \rightarrow S, S \rightarrow ASA \mid aB \mid a, A \rightarrow B \mid S \mid \epsilon, B \rightarrow b$

After removing $A \rightarrow \epsilon$, the production set becomes –

$S_0 \rightarrow S, S \rightarrow ASA \mid aB \mid a \mid AS \mid SA \mid S, A \rightarrow B \mid S, B \rightarrow b$

(3) Now we will remove the unit productions.

After removing $S \rightarrow S$, the production set becomes –

$S_0 \rightarrow S, S \rightarrow ASA \mid aB \mid a \mid AS \mid SA, A \rightarrow B \mid S, B \rightarrow b$

After removing $S_0 \rightarrow S$, the production set becomes –

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA, S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow B \mid S, B \rightarrow b$

After removing $A \rightarrow B$, the production set becomes –

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA, S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow S \mid b$

$B \rightarrow b$

After removing $A \rightarrow S$, the production set becomes –

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA, S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow b \mid ASA \mid aB \mid a \mid AS \mid SA, B \rightarrow b$

(4) Now we will find out more than two variables in the R.H.S

Here, $S_0 \rightarrow ASA, S \rightarrow ASA, A \rightarrow ASA$ violates two Non-terminals in R.H.S.

Hence we will apply step 4 and step 5 to get the following final production set which is in CNF –

$S_0 \rightarrow AX \mid aB \mid a \mid AS \mid SA$

$S \rightarrow AX \mid aB \mid a \mid AS \mid SA$

$A \rightarrow b \mid AX \mid aB \mid a \mid AS \mid SA$

$B \rightarrow b$

$X \rightarrow SA$

(5) We have to change the productions $S_0 \rightarrow aB, S \rightarrow aB, A \rightarrow aB$

And the final production set becomes –

$$S_0 \rightarrow AX \mid YB \mid a \mid AS \mid SA$$
$$S \rightarrow AX \mid YB \mid a \mid AS \mid SA$$
$$A \rightarrow b \mid A \rightarrow b \mid AX \mid YB \mid a \mid AS \mid SA$$
$$B \rightarrow b$$
$$X \rightarrow SA$$
$$Y \rightarrow a$$

Example-13 – Let us take an example to convert CFG to CNF. Consider the given grammar G1:

$$S \rightarrow ASB$$
$$A \rightarrow aAS \mid a \mid \epsilon$$
$$B \rightarrow SbS \mid A \mid bb$$

Step 1. As start symbol S appears on the RHS, we will create a new production rule $S_0 \rightarrow S$. Therefore, the grammar will become:

$$S_0 \rightarrow S$$
$$S \rightarrow ASB$$
$$A \rightarrow aAS \mid a \mid \epsilon$$
$$B \rightarrow SbS \mid A \mid bb$$

Step 2. As grammar contains null production $A \rightarrow \epsilon$, its removal from the grammar yields:

$$S_0 \rightarrow S$$
$$S \rightarrow ASB \mid SB$$
$$A \rightarrow aAS \mid aS \mid a$$
$$B \rightarrow SbS \mid A \mid \epsilon \mid bb$$

Now, it creates null production $B \rightarrow \epsilon$, its removal from the grammar yields:

$$S_0 \rightarrow S$$
$$S \rightarrow AS \mid ASB \mid SB \mid S$$
$$A \rightarrow aAS \mid aS \mid a$$
$$B \rightarrow SbS \mid A \mid bb$$

Now, it creates unit production $B \rightarrow A$, its removal from the grammar yields:

$$S_0 \rightarrow S$$
$$S \rightarrow AS \mid ASB \mid SB \mid S$$

$$A \rightarrow aAS \mid aS \mid a$$
$$B \rightarrow SbS \mid bb \mid aAS \mid aS \mid a$$

Also, removal of unit production $S_0 \rightarrow S$ from grammar yields:

$$S_0 \rightarrow AS \mid ASB \mid SB \mid S$$
$$S \rightarrow AS \mid ASB \mid SB \mid S$$
$$A \rightarrow aAS \mid aS \mid a$$
$$B \rightarrow SbS \mid bb \mid aAS \mid aS \mid a$$

Also, removal of unit production $S \rightarrow S$ and $S_0 \rightarrow S$ from grammar yields:

$$S_0 \rightarrow AS \mid ASB \mid SB$$
$$S \rightarrow AS \mid ASB \mid SB$$
$$A \rightarrow aAS \mid aS \mid a$$
$$B \rightarrow SbS \mid bb \mid aAS \mid aS \mid a$$

Step 3. In production rule $A \rightarrow aAS \mid aS$ and $B \rightarrow SbS \mid aAS \mid aS$, terminals a and b exist on RHS with non-terminates. Removing them from RHS:

$$S_0 \rightarrow AS \mid ASB \mid SB$$
$$S \rightarrow AS \mid ASB \mid SB$$
$$A \rightarrow XAS \mid XS \mid a$$
$$B \rightarrow SYS \mid bb \mid XAS \mid XS \mid a$$
$$X \rightarrow a$$
$$Y \rightarrow b$$

Also, $B \rightarrow bb$ can't be part of CNF, removing it from grammar yields:

$$S_0 \rightarrow AS \mid ASB \mid SB$$
$$S \rightarrow AS \mid ASB \mid SB$$
$$A \rightarrow XAS \mid XS \mid a$$
$$B \rightarrow SYS \mid VV \mid XAS \mid XS \mid a$$
$$X \rightarrow a$$
$$Y \rightarrow b$$
$$V \rightarrow b$$

Step 4: In production rule $S_0 \rightarrow ASB$, RHS has more than two symbols, removing it from grammar yields:

$$S_0 \rightarrow AS \mid PB \mid SB$$

$S \rightarrow AS \mid ASB \mid SB$

$A \rightarrow XAS \mid XS \mid a$

$B \rightarrow SYS \mid VV \mid XAS \mid XS \mid a$

$X \rightarrow a$

$Y \rightarrow b$

$V \rightarrow b$

$P \rightarrow AS$

Similarly, $S \rightarrow ASB$ has more than two symbols, removing it from grammar yields:

$S \rightarrow AS \mid PB \mid SB$

$S \rightarrow AS \mid QB \mid SB$

$A \rightarrow XAS \mid XS \mid a$

$B \rightarrow SYS \mid VV \mid XAS \mid XS \mid a$

$X \rightarrow a$

$Y \rightarrow b$

$V \rightarrow b$

$P \rightarrow AS$

$Q \rightarrow AS$

Similarly, $A \rightarrow XAS$ has more than two symbols, removing it from grammar yields:

$S \rightarrow AS \mid PB \mid SB$

$S \rightarrow AS \mid QB \mid SB$

$A \rightarrow RS \mid XS \mid a$

$B \rightarrow SYS \mid VV \mid XAS \mid XS \mid a$

$X \rightarrow a$

$Y \rightarrow b$

$V \rightarrow b$

$P \rightarrow AS$

$Q \rightarrow AS$

$R \rightarrow XA$

Similarly, $B \rightarrow SYS$ has more than two symbols, removing it from grammar yields:

$$S \rightarrow AS \mid PB \mid SB$$
$$S \rightarrow AS \mid QB \mid SB$$
$$A \rightarrow RS \mid XS \mid a$$
$$B \rightarrow TS \mid VV \mid XAS \mid XS \mid a$$
$$X \rightarrow a$$
$$Y \rightarrow b$$
$$V \rightarrow b$$
$$P \rightarrow AS$$
$$Q \rightarrow AS$$
$$R \rightarrow XA$$
$$T \rightarrow SY$$

Similarly, $B \rightarrow XAX$ has more than two symbols, removing it from grammar yields:

$$S \rightarrow AS \mid PB \mid SB$$
$$S \rightarrow AS \mid QB \mid SB$$
$$A \rightarrow RS \mid XS \mid a$$
$$B \rightarrow TS \mid VV \mid US \mid XS \mid a$$
$$X \rightarrow a$$
$$Y \rightarrow b$$
$$V \rightarrow b$$
$$P \rightarrow AS$$
$$Q \rightarrow AS$$
$$R \rightarrow XA$$
$$T \rightarrow SY$$
$$U \rightarrow XA$$

PRACTICE PROBLEMS BASED ON CHOMSKY NORMAL FORM-

Problem-01:

Convert the given grammar to CNF-

$$\begin{aligned}S &\rightarrow aAD \\ A &\rightarrow aB / bAB \\ B &\rightarrow b \\ D &\rightarrow d\end{aligned}$$

Solution-

Step-01:

The given grammar is already completely reduced.

Step-02:

The productions already in chomsky normal form are-

$$B \rightarrow b \dots\dots\dots(1)$$

$$D \rightarrow d \dots\dots\dots(2)$$

These productions will remain as they are.

The productions not in chomsky normal form are-

$$S \rightarrow aAD \dots\dots\dots(3)$$

$$A \rightarrow aB / bAB \dots\dots\dots(4)$$

We will convert these productions in chomsky normal form.

Step-03:

Replace the terminal symbols a and b by new variables C_a and C_b .

This is done by introducing the following two new productions in the grammar-

$$C_a \rightarrow a \dots\dots\dots(5)$$

$$C_b \rightarrow b \dots\dots\dots(6)$$

Now, the productions (3) and (4) modifies to-

$$S \rightarrow C_aAD \dots\dots\dots(7)$$

$$A \rightarrow C_aB / C_bAB \dots\dots\dots(8)$$

Step-04:

Replace AD and AB by new variables C_{AD} and C_{AB} respectively.

This is done by introducing the following two new productions in the grammar-

$$C_{AD} \rightarrow AD \dots\dots\dots(9)$$

$$C_{AB} \rightarrow AB \dots\dots\dots(10)$$

Now, the productions (7) and (8) modifies to-

$$S \rightarrow C_aC_{AD} \dots\dots\dots(11)$$

$$A \rightarrow C_aB / C_bC_{AB} \dots\dots\dots(12)$$

Step-05:

From (1), (2), (5), (6), (9), (10), (11) and (12), the resultant grammar is-

$$S \rightarrow C_aC_{AD}$$

$$A \rightarrow C_aB / C_bC_{AB}$$

$$B \rightarrow b$$

$D \rightarrow d$

$C_a \rightarrow a$

$C_b \rightarrow b$

$C_{AD} \rightarrow AD$

$C_{AB} \rightarrow AB$

This grammar is in chomsky normal form.

Problem-02:

Convert the given grammar to CNF-

$S \rightarrow 1A / 0B$

$A \rightarrow 1AA / 0S / 0$

$B \rightarrow 0BB / 1S / 1$

Solution-

Step-01:

The given grammar is already completely reduced.

Step-02:

The productions already in chomsky normal form are-

$A \rightarrow 0 \dots\dots\dots(1)$

$B \rightarrow 1 \dots\dots\dots(2)$

These productions will remain as they are.

The productions not in chomsky normal form are-

$$S \rightarrow 1A / 0B \dots\dots\dots(3)$$

$$A \rightarrow 1AA / 0S \dots\dots\dots(4)$$

$$B \rightarrow 0BB / 1S \dots\dots\dots(5)$$

We will convert these productions in chomsky normal form.

Step-03:

Replace the terminal symbols 0 and 1 by new variables C and D.

This is done by introducing the following two new productions in the grammar-

$$C \rightarrow 0 \dots\dots\dots(6)$$

$$D \rightarrow 1 \dots\dots\dots(7)$$

Now, the productions (3), (4) and (5) modifies to-

$$S \rightarrow DA / CB \dots\dots\dots(8)$$

$$A \rightarrow DAA / CS \dots\dots\dots(9)$$

$$B \rightarrow CBB / DS \dots\dots\dots(10)$$

Step-04:

Out of (8), (9) and (10), the productions already in Chomsky Normal Form are-

$$S \rightarrow DA / CB \dots\dots\dots(11)$$

$$A \rightarrow CS \dots\dots\dots(12)$$

$$B \rightarrow DS \dots\dots\dots(13)$$

These productions will remain as they are.

The productions not in chomsky normal form are-

$$A \rightarrow DAA \dots\dots\dots(14)$$

$$B \rightarrow CBB \dots\dots\dots(15)$$

We will convert these productions in Chomsky Normal Form.

Step-05:

Replace AA and BB by new variables E and F respectively.

This is done by introducing the following two new productions in the grammar-

$$E \rightarrow AA \dots\dots\dots(16)$$
$$F \rightarrow BB \dots\dots\dots(17)$$

Now, the productions (14) and (15) modifies to-

$$A \rightarrow DE \dots\dots\dots(18)$$
$$B \rightarrow CF \dots\dots\dots(19)$$

Step-06:

From (1), (2), (6), (7), (11), (12), (13), (16), (17), (18) and (19), the resultant grammar is-

$$S \rightarrow DA / CB$$
$$A \rightarrow CS / DE / 0$$
$$B \rightarrow DS / CF / 1$$
$$C \rightarrow 0$$
$$D \rightarrow 1$$
$$E \rightarrow AA$$
$$F \rightarrow BB$$

This grammar is in chomsky normal form.

To gain better understanding about Chomsky Normal Form,

